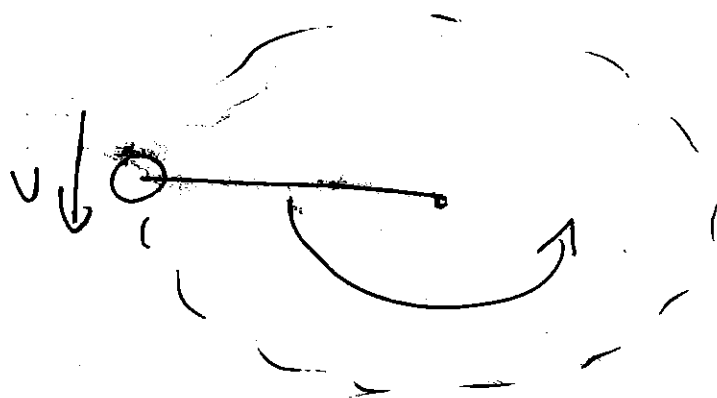


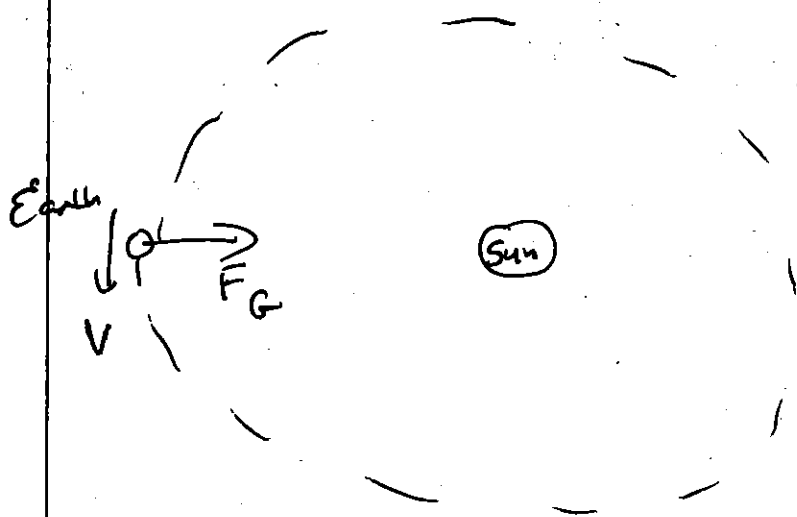
6 Circular motion

Ball on string spins around



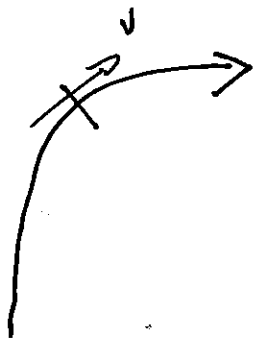
$$\vec{O} \rightarrow T$$

$$\frac{mv^2}{r}$$



$$\vec{O} \rightarrow F_G$$

$$\frac{mv^2}{r}$$

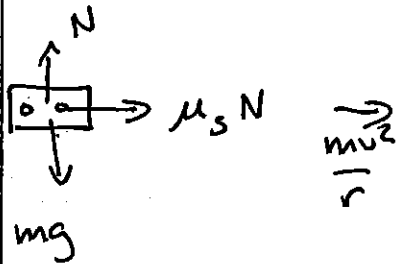


Car goes around a
flat curve on ice

$$\mu_s = 0.1$$

$$V_{\max} = ?$$

$$r = 50 \text{ m}$$



$$\sum F_y: N - mg = 0$$

$$(N = mg)$$

$$\sum F_x: \mu_s mg = \frac{mv^2}{r}$$

$$F_f = \mu_s N \quad \text{max it can be!}$$

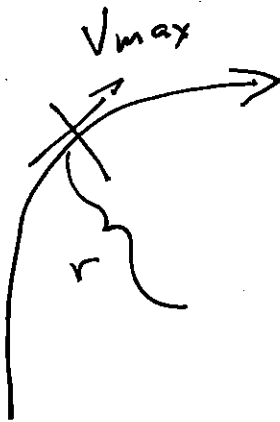
$$V_{\max} = \sqrt{\mu_s g r}$$

$$V_{\max} = \sqrt{0.1 \cdot 9.8 \cdot 50}$$

$$= 7 \text{ m/s} \quad (15 \text{ mph})$$

Banked curve

3/7

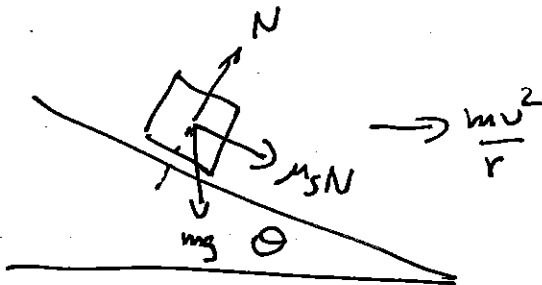


$$\theta = 10^\circ$$

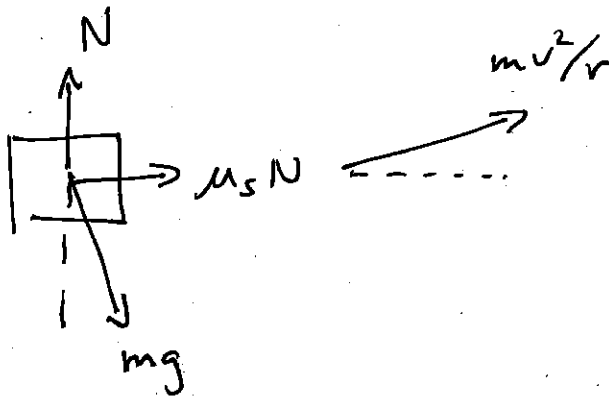
$$\mu_s = 0.3$$

$$r = 50 \text{ m}$$

$$v_{\max} = ?$$



no mg!



$$\Sigma F_y: N - mg \cos \theta = \frac{mv^2}{r} \sin \theta$$

$$N = mg \cos \theta + \frac{mv^2}{r} \sin \theta$$

$$\Sigma F_x: mg \sin \theta + \mu_s N = \frac{mv^2}{r} \cos \theta$$

$$mg \sin \theta + \mu_s mg \cos \theta + \mu_s \frac{mv^2}{r} \sin \theta = \frac{mv^2}{r} \cos \theta$$

cancel m divide by $\cos \theta$ multiply by r

$$g \sin \theta + \mu_s g \cos \theta + \mu_s \frac{v^2}{r} \sin \theta = \frac{v^2}{r} \cos \theta \quad 4/7$$

$$g \tan \theta + \mu_s g + \frac{\mu_s v^2}{r} \tan \theta = \frac{v^2}{r}$$

$$gr \tan \theta + \mu_s gr + \mu_s v^2 \tan \theta = v^2 \quad \text{save eq.}$$

for $v_{\max}$
on banked curve

$$v_{\max} = ?$$

$$gr \tan \theta + \mu_s gr = v^2 - \mu_s v^2 \tan \theta$$

$$gr (\tan \theta + \mu_s) = v^2 (1 - \mu_s \tan \theta)$$

$$v^2 = gr \frac{\mu_s + \tan \theta}{1 - \mu_s \tan \theta}$$

$$v_{\max} = \sqrt{gr \frac{\mu_s + \tan \theta}{1 - \mu_s \tan \theta}}$$

$$r = 50 \text{ m} \quad \mu_s = 0.1 \quad \theta = 10^\circ$$

$$\tan 10^\circ = 0.17632698$$

$$v_{\max} = \sqrt{9.8 \frac{\text{m}}{\text{s}^2} 50 \text{ m} \frac{0.1 + 0.176}{1 - 0.1 \cdot 0.176}}$$

$$v_{\max} = \sqrt{9.8 \cdot 50 \frac{0.276}{0.982}} = 11.7 \text{ m/s} \quad (26 \text{ mph})$$

$$\mu_s = 0.2$$

$$V_{max} = \sqrt{9.8 \cdot 50 \cdot \frac{0.2 + 0.176}{1 - 0.2 \cdot 0.176}} \quad 5/7$$

$$V_{max} = 13.8 \text{ m/s} \quad 31 \text{ mph}$$

$$\mu_s = 0.3$$

$$V_{max} = 15.7 \text{ m/s} \quad 35 \text{ mph}$$

$$\mu_s = 0.4$$

$$V_{max} = 17.4 \text{ m/s} \quad 39 \text{ mph}$$

why so slow $r = 50 \text{ m}$ is tight

$\theta = 10^\circ$ is small

if $r = 100 \text{ m}$ (cloverleaf)

$$\theta = 20^\circ$$

$$\tan 20^\circ = 0.364$$

$$\mu = 0.3$$

$$V_{max} = \sqrt{9 \cdot r \cdot \frac{\mu_s + \tan \theta}{1 - \mu_s \tan \theta}}$$

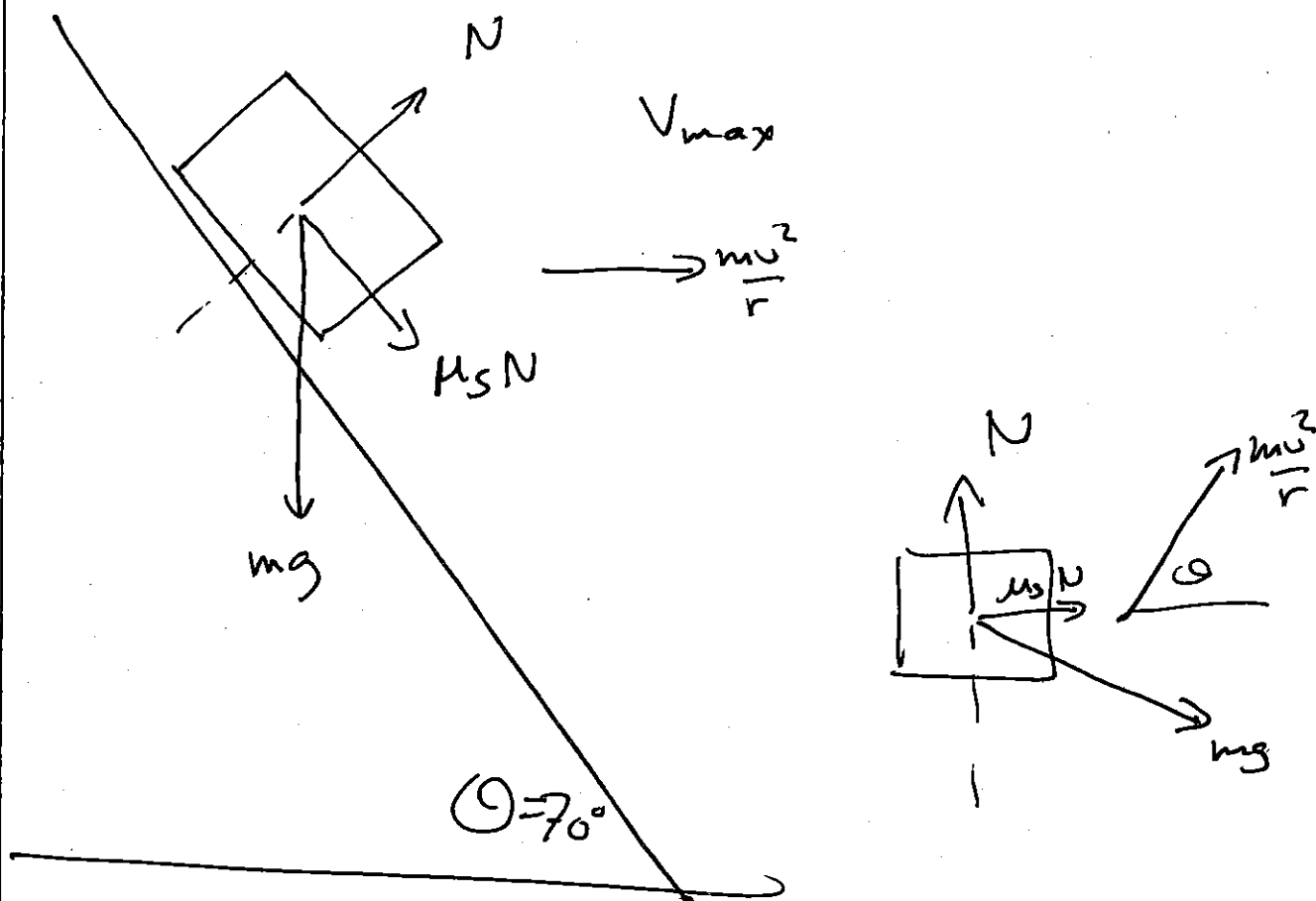
$$= 27 \text{ m/s} \quad (60 \text{ mph})$$

6/7

What happens if θ is large $\sim 70^\circ$

μ_s is small $\mu_s \approx 0.2$ $r = 60\text{m}$?

$v_{\max}$ we understand. What about $v_{\min}$?



$$\Sigma F_y: N - mg \cos \theta = m \frac{v^2}{r} \sin \theta$$

$$N = mg \cos \theta + \frac{mv^2}{r} \sin \theta$$

$$\Sigma F_x: mg \sin \theta + \mu_s N = \frac{mv^2}{r} \cos \theta$$

$$mg \sin \theta + \mu_s mg \cos \theta + \mu_s \frac{mv^2}{r} \sin \theta = \frac{mv^2}{r} \cos \theta$$

$$\frac{r}{m \cos \theta}$$

$$g r \tan \theta + \mu g r + \mu v^2 \tan \theta = v^2 \quad \frac{7}{7}$$

$$v^2 (1 - \mu \tan \theta) = g r (\mu + \tan \theta)$$

$$V_{\max} = \sqrt{g r \frac{\mu + \tan \theta}{1 - \mu \tan \theta}}$$

$$\tan 70^\circ = 2.75$$

$$V_{\max} = \sqrt{9.8 \cdot 60 \cdot \frac{0.2 + 2.75}{1 - 0.2 \cdot 2.75}}$$

$$= \sqrt{588 \cdot \frac{2.95}{0.45}} = 62 \text{ m/s } (138 \text{ mph})$$

$$V_{\min} \quad g r \tan \theta - \mu g r - \mu v^2 \tan \theta = v^2$$

$$v^2 (1 + \mu \tan \theta) = g r (\tan \theta - \mu)$$

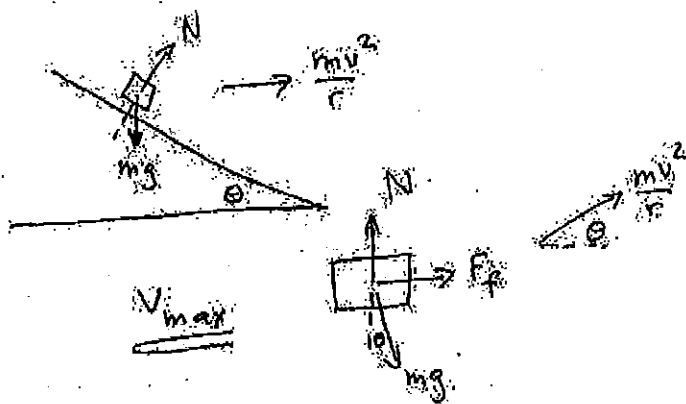
$$V_{\min} = \sqrt{g r \frac{\tan \theta - \mu}{1 + \mu \tan \theta}}$$

$$= \sqrt{9.8 \cdot 60 \cdot \frac{2.75 - 0.2}{1 + 0.2 \cdot 2.75}}$$

$$= \sqrt{588 \cdot \frac{2.55}{1.55}} = 31 \text{ m/s } (69 \text{ mph})$$

1/3

There is a banked turn $\mu = 0.2$ $\theta = 40^\circ$.
 The radius of the curve is 70 m. What is
 the maximum speed to take the corner at?
 What is the minimum speed?



$$N = mg \cos \theta = \frac{mv^2}{r} \sin \theta$$

$$N = mg \cos \theta + \frac{mv^2}{r} \sin \theta$$

$$mg \sin \theta + F_f = \frac{mv^2}{r} \cos \theta$$

$$mg \sin \theta + \mu mg \cos \theta + \mu \frac{mv^2}{r} \sin \theta = \frac{mv^2}{r} \cos \theta$$

$$rg \sin \theta + \mu rg \cos \theta + \mu v^2 \sin \theta = v^2 \cos \theta$$

$$v^2 (\cos \theta - \mu \sin \theta) = gr (\sin \theta + \mu \cos \theta)$$

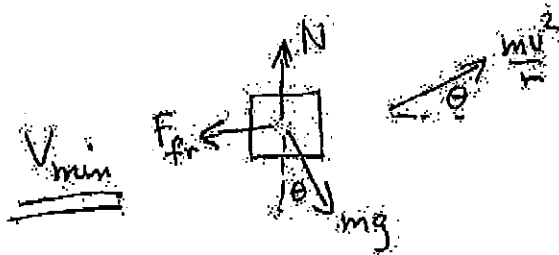
$$v = \sqrt{gr \left(\frac{\sin \theta + \mu \cos \theta}{\cos \theta - \mu \sin \theta} \right)}$$

$$V_{\max} = \sqrt{gr \frac{\sin \theta + \mu \cos \theta}{\cos \theta - \mu \sin \theta}}$$

$$= \sqrt{9.8 \frac{\text{m}}{\text{s}^2} 70 \text{ m} \frac{\sin 40^\circ + 0.2 \cos 40^\circ}{\cos 40^\circ - 0.2 \sin 40^\circ}}$$

2/3

$$V_{\max} = 29.3 \text{ m/s}$$



$$y: N - mg \cos \theta = \frac{mv^2}{r} \sin \theta$$

$$N = mg \cos \theta + \frac{mv^2}{r} \sin \theta$$

$$x: mg \sin \theta - F_{fr} = \frac{mv^2}{r} \cos \theta$$

$$mg \sin \theta - \mu mg \cos \theta - \mu \frac{mv^2}{r} \sin \theta = \frac{mv^2}{r} \cos \theta$$

$$rg \sin \theta - \mu rg \cos \theta - \mu v^2 \sin \theta = v^2 \cos \theta$$

$$v^2 (\cos \theta + \mu \sin \theta) = rg (\sin \theta - \mu \cos \theta)$$

$$v = \sqrt{gr \left(\frac{\sin \theta - \mu \cos \theta}{\cos \theta + \mu \sin \theta} \right)}$$

$$V_{min} = \sqrt{9.8 \frac{m}{s^2} 70m \left(\frac{\sin 40^\circ - 0.2 \cos 40^\circ}{\cos 40^\circ + 0.2 \sin 40^\circ} \right)} \quad 3/3$$

$$V_{min} = 19.4 \frac{m}{s}$$

$$V_{max} = 66 \text{ mph}$$

$$V_{min} = 44 \text{ mph}$$

seems pretty familiar doesn't it?